Real-Time Climate Crisis Dashboard (RTCCD)

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Our Objective

U2 Real-Time Climate Crisis Dashboard

Research Area: Data Science · Data Visualization

Rationale: *Record-breaking wildfires, floods, and heatwaves in 2025 exposed gaps in public access to real-time environmental data.*

Challenge: Develop an interactive dashboard that aggregates publicly available real-time data on wildfires, air quality (AQI), flood alerts, and temperature anomalies. The dashboard should allow geographic filtering and alert users when critical thresholds are exceeded in their region. You may use open APIs such as OpenAQ, NASA FIRMS, and NOAA.

How We Did It

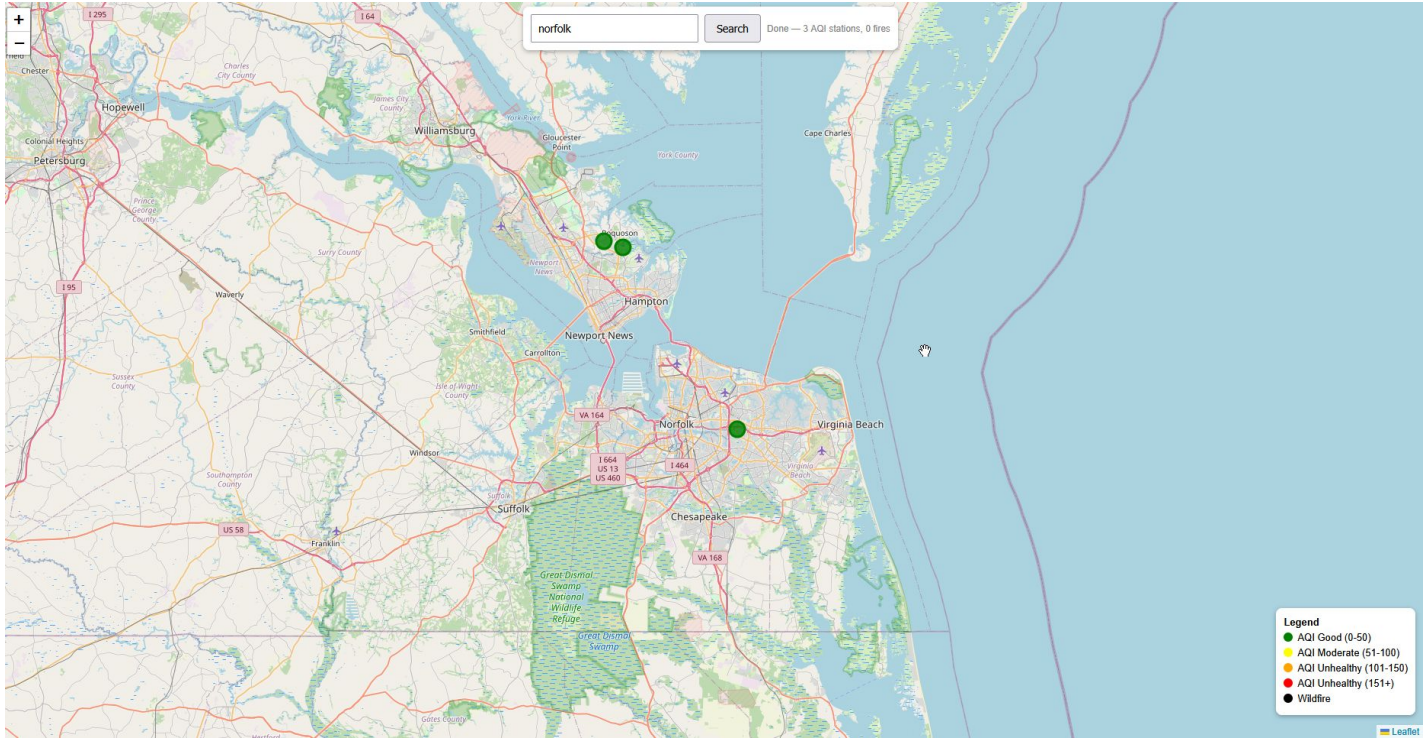
rtccd > src > main > java > com > csgs > firms > firmsClient.java

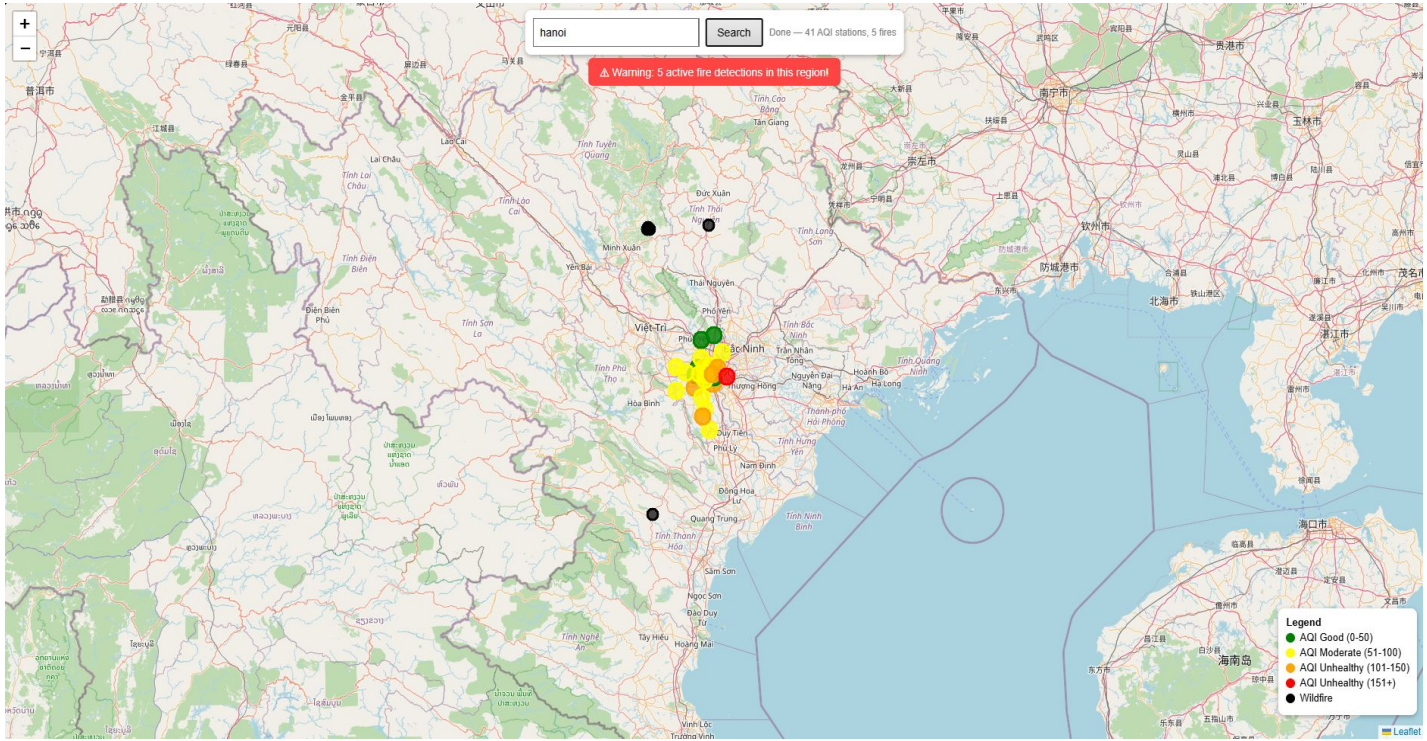
```java
public class firmsClient {
    private Dotenv dotenv = Dotenv.load();
    private final String MAP_KEY = dotenv.get("FIRMS_KEY");
private final String BASE_URL =
    "https://firms.modaps.eosdis.nasa.gov/api/area/csv/"
    + MAP_KEY +
    "/MODIS_NRT/116.542969,-34.741612,146.777344,-17.811456/1";
    private final HttpClient client;

    public firmsClient(){
        client = HttpClient.newHttpClient();
    }

  public String findAll(Coordinates coords) throws IOException, InterruptedExcep
    double[] bbox = coords.boundingBox();
    String url = "https://firms.modaps.eosdis.nasa.gov/api/area/csv/"
        + MAP_KEY
        + "/MODIS_NRT/"
        + bbox[1] + "," + bbox[0] + "," + bbox[3] + "," + bbox[2]
        + "/1";

    System.out.println("FIRMS URL: " + url); // debug

    HttpRequest request = HttpRequest.newBuilder()
        .uri(URI.create(url))
        .timeout(Duration.ofSeconds(10)) // give up after 10 seconds
        .GET()
        .build();

    HttpResponse<String> response = client.send(request, HttpResponse.BodyHandle
    System.out.println("FIRMS status: " + response.statusCode());
    System.out.println("FIRMS raw: " + response.body().substring(0, Math.min(200
    return response.body();
}

    public String findAll(double[] bbox) throws IOException, InterruptedExceptio
    String url = "https://firms.modaps.eosdis.nasa.gov/api/area/csv/"
        + MAP_KEY +
        "/MODIS_NRT/"
        + bbox[1] + "," + bbox[0] + "," + bbox[3] + "," + bbox[2]
        + "/1";
```

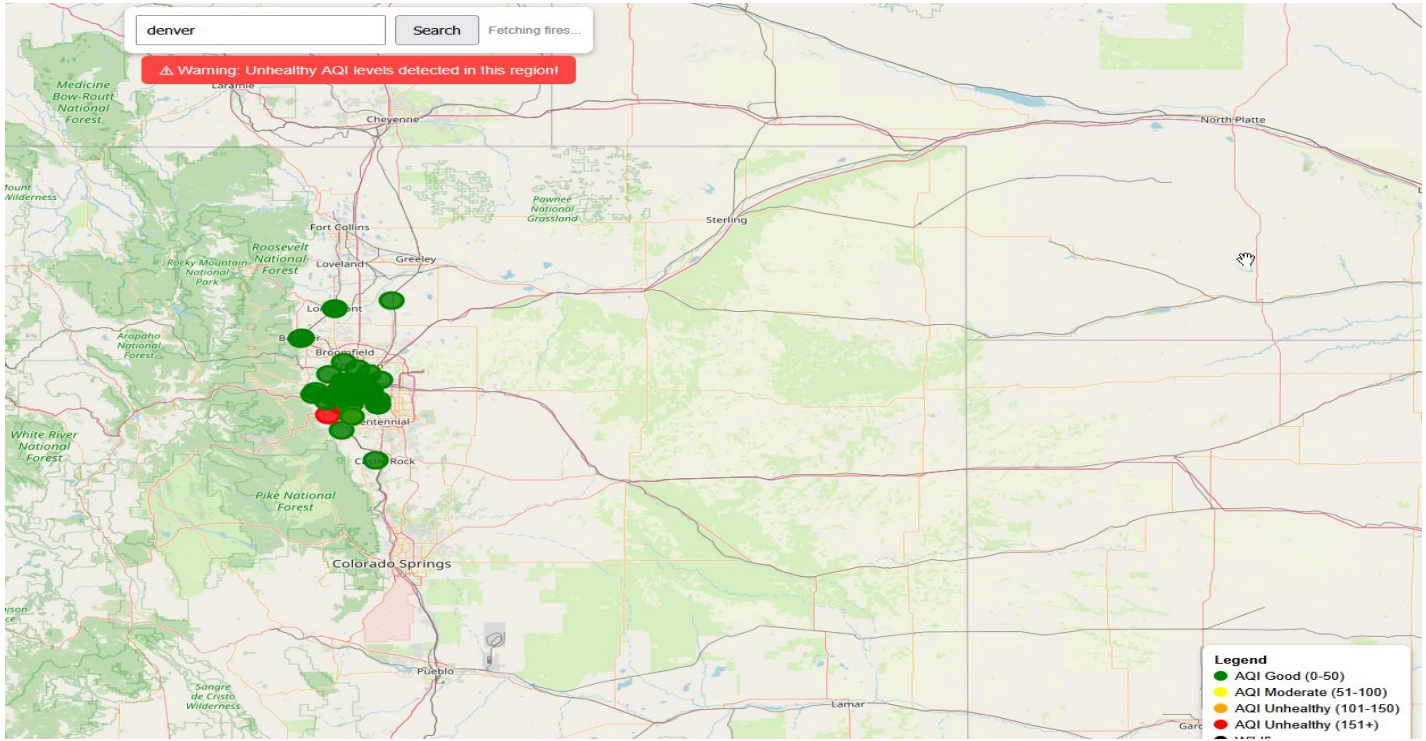
rtccd > src > main > java > com > csgs > openAQ > openAQClient.java > openAQClient > parselocationJson(String)

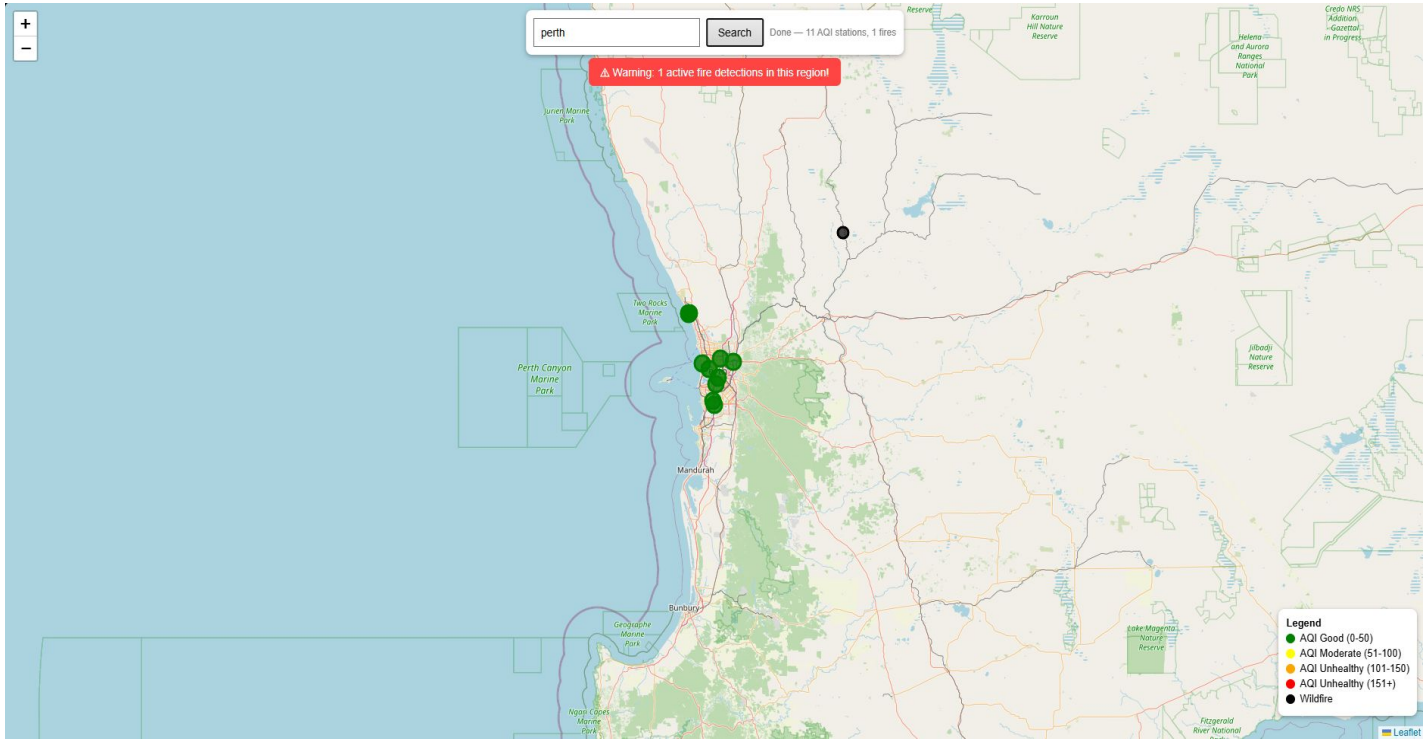
```java
public class openAQClient {

    private Dotenv dotenv = Dotenv.load();

    private final String APIKEY = dotenv.get("OPENAQ_KEY");
    private final String BASEURL = "https://api.openaq.org/v3/locations?bbox=-11
    private final HttpClient client;

    public openAQClient() {
        client = HttpClient.newHttpClient();
    }

    public String findAll() throws IOException, InterruptedException {
        HttpRequest request = HttpRequest.newBuilder()
                .uri(URI.create(BASEURL))
                .header("x-api-key", APIKEY)
                .GET()
                .build();
        HttpResponse<String> response = client.send(request, HttpResponse.BodyHa
        return response.body();
    }

    public String findAll(Coordinates coords) throws IOException, InterruptedExc
        double[] bbox = coords.boundingBox();
        String url = "https://api.openaq.org/v3/locations?bbox="
                + bbox[1] + "," + bbox[0] + "," + bbox[3] + "," + bbox[2];
        //System.out.println("URL: " + url);
        HttpRequest request = HttpRequest.newBuilder()
                .uri(URI.create(url))
                .header("x-api-key", APIKEY)
                .GET()
                .build();
        HttpResponse<String> response = client.send(request, HttpResponse.BodyHa
        return response.body();
    }

    // bbox = {west, south, east, north}
    public String findAll(double[] bbox) throws IOException, InterruptedExceptio
        String url = "https://api.openaq.org/v3/locations?bbox="
                + bbox[0] + "," + bbox[1] + "," + bbox[2] + "," + bbox[3];
        HttpRequest request = HttpRequest.newBuilder()
```

The Finished Product




hanoi
Search
Done — 41 AQI stations, 5 fires
⚠ Warning: 5 active fire detections in this region!
Legend
AQI Good (0-50)
AQI Moderate (51-100)
AQI Unhealthy (101-150)
AQI Unhealthy (151+)
Wildfire
Leaflet


denver
Search
Fetching fires...
⚠ Warning: Unhealthy AQI levels detected in this region!
Medicine Bow-Routt National Forest
Laramie
Cheyenne
North Platte
Pawnee National Grassland
Sterling
Fort Collins
Roosevelt National Forest
Rocky Mountain National Park
Loveland
Greeley
Arapaho National Forest
Broomfield
White River National Forest
Pike National Forest
Colorado Springs
Pueblo
Sangre de Cristo Wilderness
Lamar
Legend
AQI Good (0-50)
AQI Moderate (51-100)
AQI Unhealthy (101-150)
AQI Unhealthy (151+)


perth
Search
Done — 11 AQI stations, 1 fires
⚠ Warning: 1 active fire detections in this region!
Karroun Hill Nature Reserve
Helena and Aurora Ranges National Park
Credo NRS Addition in Gazettal in Progress
Jurien Marine Park
Two Rocks Marine Park
Perth Canyon Marine Park
Jilbadji Nature Reserve
Mandurah
Bunbury
Geographe Marine Park
Ngari Capes Marine Park
Lake Magenta Nature Reserve
Fitzgerald River National
Legend
AQI Good (0-50)
AQI Moderate (51-100)
AQI Unhealthy (101-150)
AQI Unhealthy (151+)
Wildfire
Leaflet

Reflection

- What did we learn?
- What could we improve on?
- What can you use this program for?